

Hearables: Automatic Sleep Scoring from Single-Channel Ear-EEG in Older Adults

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Abstract—Sleep disorders are a prevalent problem among older adults, yet obtaining an accurate and reliable assessment of sleep quality can be challenging. Traditional polysomnography (PSG) is the gold standard for sleep staging, but is obtrusive, expensive, and requires expert assistance. To this end, we propose a minimally invasive single-channel single ear-EEG automatic sleep staging method for older adults. The method employs features from the frequency, time, and structural complexity domains, which provide a robust classification of sleep stages from a standardised viscoelastic earpiece. Our method is verified on a dataset of older adults and achieves a kappa value of at least 0.61, indicating substantial agreement. This paves the way for a non-invasive, cost-effective, and portable alternative to traditional PSG for sleep staging.

I. INTRODUCTION

Sleep monitoring is an aspect of geriatric care as it allows for the identification and management of sleep disorders in the older adult population [1]. As individuals age, changes in physiological and psychological processes can lead to an increased prevalence of sleep disorders, such as insomnia and sleep apnea. These disorders can have a significant impact on the health and well-being of the elderly, as they are associated with an increased risk of falls, cognitive decline, and cardiovascular disease [2]. Studies have shown that implementing proper treatment for sleep disorders can improve sleep quality and overall health outcomes in older adults [3]. Sleep monitoring at home is therefore a prerequisite for objective measures of sleep patterns, allowing for the accurate diagnosis and monitoring of sleep disorders. In addition, such monitoring can also be used to track the effectiveness of any interventions or treatments implemented, providing valuable information for optimized care.

Hearables [4], have made it possible to track multiple physiological signals such as the electroencephalogram (EEG) [5], [6], electrocardiogram (ECG) [7]–[9], and even cognitive workload [10] and daily activities [11]. The application of Hearables to sleep has been a very active area of research [12]–[16]. Results from studies using a standardized and generic in-ear EEG sensor [12], [13] show substantial agreement between automatic sleep stage prediction based on

ear-EEG and the hypnogram based on the full polysomnography (PSG), the gold standard for monitoring sleep and evaluating patients' sleep disorders in a clinic. However, the major issue with these sleep studies is that they are typically conducted on young, healthy adults. Given the prevalence of sleep disorders in the elderly population and their weaker EEG, this study sets out to examine the utility of hearables for monitoring sleep in the elderly. We demonstrate that the use of hearables to track physiological signals makes it possible to gain a better understanding, leading to effective monitoring and managing of sleep disorders in older adults.

II. METHODS

A. Experimental Setup and Protocol

This study involved 17 participants (71.8 ± 4.4 years (mean $\pm$ SD); range: 65–83 years; 6 women) with no self-reported history of mental health or neurological problems. Participants had self-reported comorbidities, including type-2 diabetes, sleep apnea, hypertension, etc., but these were reported to be stable and well-controlled, with no recent medication changes or hospitalizations. Participants were invited to attend an overnight residential session at the Surrey Sleep Research Centre (SSRC). This included a 10-hour in-bed period during which polysomnography (PSG) was recorded in accordance with American Academy of Sleep Medicine (AASM) guidelines [17] using the SomnoHD recording system and DOMINO software (Somnomedics, Germany). A standard AASM adult PSG montage was used and in-ear EEG sensor [12], as depicted in Fig. 1. Data from one ear were collected simultaneously using the same recording system, so as to synchronize with scalp EEG. Sleep stages were scored at 30-second epochs by two independent scorers, and a consensus hypnogram was generated. The study was conducted in accordance with the principles of the Declaration of Helsinki and Good Clinical Practice. The protocol received a favourable opinion from the University of Surrey Ethics Committee and potential participants were provided with detailed information about the study protocol, and they gave a written informed consent.

B. Data Pre-Processing

Data pre-processing is a crucial step in which the recorded ear and scalp EEG data is cleaned, filtered, and prepared for analysis. The recorded ear and scalp EEG were sampled at 256 Hz, referenced to the contralateral mastoid and pre-processed as follows:

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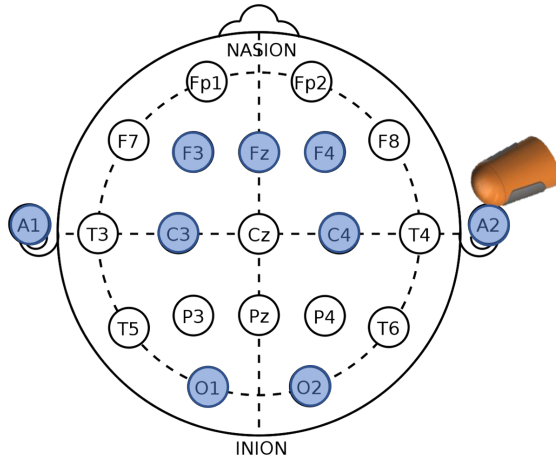


Fig. 1. The experimental EEG setup, with the scalp electrodes and the generic Ear-EEG sensor visible at A2.

- Filtering by a 1-30 Hz, 8th order Butterworth band-pass filter, to remove baseline drifts and higher frequency bands in EEG that are not related to sleep;
- The variance of each 30-sec epoch was calculated. If the value of the variance was greater than three standard deviations from the median of that epoch, the epoch was removed from the dataset.

Fig. 2 shows a 30-sec epoch of a single ear-EEG channel and the C4-M1 scalp channel along with their corresponding power spectral density (PSD) plots. Observe the similarity between the scalp EEG and ear-EEG during the wake (top panel) and N3 sleep (bottom panel) stages. This is also reflected in the corresponding power spectra (rightmost panels), where during the wake epoch both the scalp and ear channel had a peak at 10 Hz. Similarly, in the epoch of N3, the dominant frequency was the delta band between 0.5 Hz and 4 Hz, visible in both the ear and the scalp EEG.

C. Feature Extraction

We aimed to extract features that have been commonly used for automatic sleep staging, especially when EEG is recorded from inside the ear canal. The features spanned the time domain, frequency domain and structural complexity domain of the signals; in total 41 features were used.

1) *Time domain*: The Hjorth parameters are a set of mathematical descriptors that can be used to quantify the complexity, activity, and mobility of a signal, and are given by $Activity = \sqrt{\frac{var(dy(t)/dt)}{var(y(t))}}$, $Mobility = \frac{Mobility(dy(t)/dt)}{Mobility(y(t))}$, where $y(t)$ is the signal. Physically, the activity is a measure of the overall power of the signal, while the mobility is a measure of the frequency content of the signal, and the complexity is a measure of the variability of the signal. These parameters have been used in several studies [14], [16], [18] as features for automatic sleep staging and have been shown to be effective in differentiating between stages of sleep, such as REM and non-REM sleep.

2) *Frequency domain*: The Spectral Edge Frequency (SEF) and the relative band powers of the alpha, beta, theta, and delta bands in EEG were calculated from the power spectral density (PSD) of each 30-second epoch using Welch's averaged periodogram with the window length of 4 sec. The SEF is a frequency domain feature which is defined as the frequency value that contains a certain percentage of the power within a given frequency range. It is commonly used in physiological data analysis, especially in studies of EEG [12], [13], [19]. The chosen frequency ranges were SEF50, SEF95, and SEFd (which is SEF95-SEF50) in the following bands; $\delta - \beta = 0.5-30$ Hz, $\delta - \alpha = 0.5-16$ Hz, $\theta = 2-8$ Hz, $\alpha = 8-15$ Hz, $\alpha_l = 8-11$ Hz, $\alpha_h = 11-15$ Hz, and $\beta = 16-30$ Hz.

In addition to the SEF, the relative band powers of the alpha (8-12 Hz), beta (13-30 Hz), theta (4-8 Hz) and delta (0.5-4 Hz) bands along with the total power were also used as features. These frequency bands are commonly associated with different stages of sleep, such as delta waves with deep sleep, theta waves with light sleep and drowsiness, alpha waves with relaxation and beta waves with wakefulness.

3) *Structural complexity*: The Multi-Scale Entropy (MSE) technique is used to evaluate the structural complexity of time-series data across multiple time scales. This method can be implemented using various algorithms such as sample entropy, approximate entropy, and permutation entropy [19]. In our study, we employed the Multi-Scale Fuzzy Entropy (MSFE) method [13], which is particularly effective in handling noise. The specific parameters chosen for MSFE in this study were: a maximum scale $\tau = 15$, $m = 2$, $n = 2$, $r = 0.15 \times (\text{standard deviation of each epoch})$; this choice is in accordance with our previous study using the same in-ear sensor [13].

D. Training of Classifiers and Evaluation of Predictions

We evaluated the performance of 13 different supervised classifiers for automatic sleep staging, listed in Fig. 3. The stratified sampling approach was used as it conserves class proportions across the training and the test data, thus reducing model bias and improving accuracy. A proportion of 70% of the data was used for training and validation and 30% was held out as the unseen test set. Ten-fold cross-validation was employed on the stratified training data, and the classifier with the highest kappa value was chosen. Cohen's kappa value can be considered as a measure of agreement between the predicted sleep stages, and the reference sleep stages obtained from the consensus hypnogram as ground truth labels. Along with the kappa value, the sensitivity, precision and accuracy were also calculated [12].

III. RESULTS AND DISCUSSION

This study employs a single-channel ear-EEG sensor for automatic sleep scoring in an elderly population with a mean age of around 72 years. The kappa value was used as a metric based on which the choice of the model was made. Among the different types of classifiers shown in Fig. 3, the ensemble learning classifiers performed the best with the highest kappa

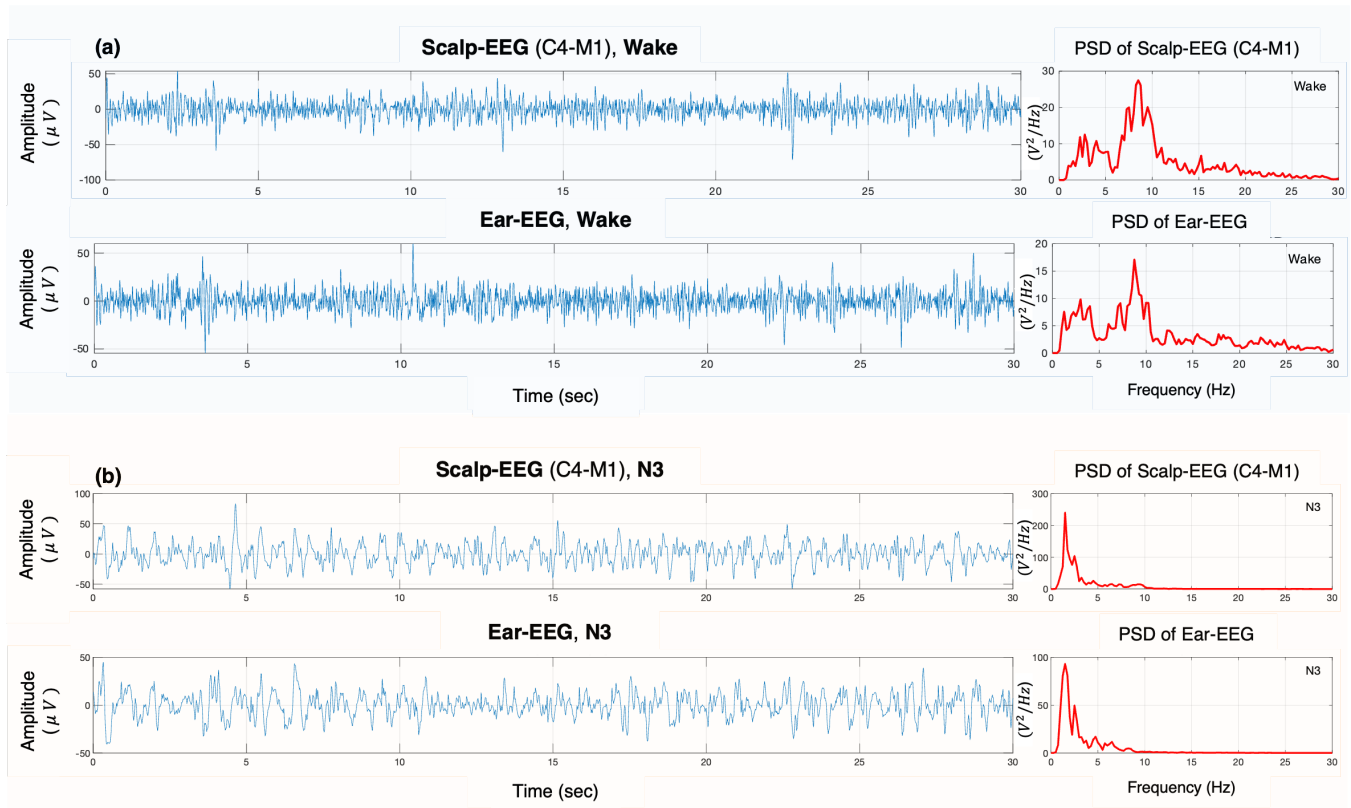


Fig. 2. Sample segments of the pre-processed scalp and in-ear EEG from (a) the wake and (b) the N3 sleep stage with their corresponding power-spectral density plots (rightmost).

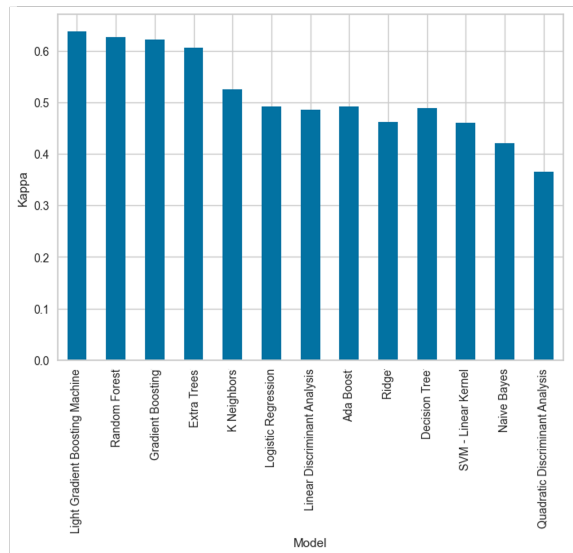


Fig. 3. Kappa values of the classifiers used.

attained from the Light Gradient Boosting Machine model (LGBM). Ensemble classifiers combine multiple individual models to achieve improved accuracy, reduced overfitting, increased robustness, higher diversity, and better handling of complex decision boundaries. The test results of the LGBM model, shown in the confusion matrix in Fig. 4 indicate the highest sensitivity for the wake (90.6%), N2 (83.6%) and

N3 (75.6%) stages, while the least sensitivity of 12.4% was observed for the N1 and REM (41.3%) stages. These results are consistent with the literature [14], [15] as N1 stage has the lowest proportion of data and is hence not sufficiently represented in the training set, while REM spans a range of frequencies with high and low components making it harder to detect. Fig. 5 shows the classification results per participant, with the kappa value for 75% of the subjects at 0.61, thus indicating substantial agreement. Compared to previous studies that used the same in-ear sensor for young and healthy adults [12], [13], the classification accuracy and kappa values were very similar, despite the generally weaker EEG in older adults. More specifically, the overall average accuracy in this study was around 74%, with a kappa value of 0.63 which indicates substantial agreement between the predicted and true values that were obtained from the gold standard, manually scored hypnogram. When casting the 5-class problem into 4, 3, and 2 classes by combining some classes and retraining the model, the performance increasingly and considerably improved. For example, by just combining N1 and N2 stages into one class, the overall average accuracy and kappa value increased respectively to 81% and 0.71, as shown in Table I. Observe from Table I that as more classes are aggregated together, the sensitivity and precision increased, except for REM, where the values were constantly around 68% and 42% for the precision and sensitivity, respectively. This indicates a need for different

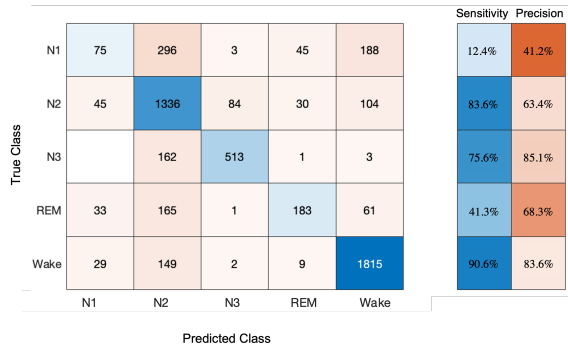


Fig. 4. Confusion matrix for the unseen test data in the five-stage prediction task.

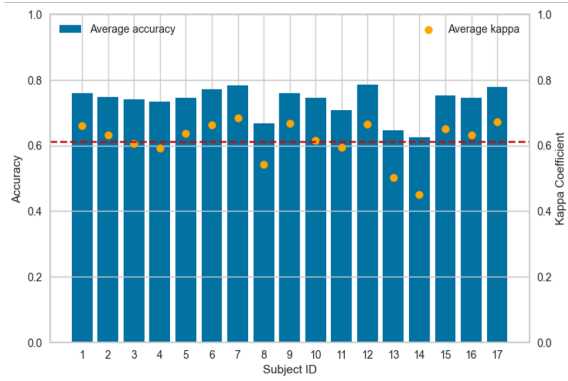


Fig. 5. Classification accuracy (blue bars) and kappa values (orange dots) for the 17 individual participants for a 5-class sleep staging task.

features to be designed specifically to capture REM and hence increase the performance of the classifier.

TABLE I
PRECISION (SENSITIVITY) FOR SLEEP CLASSIFICATION IN DIFFERENT CLASSIFYING SCENARIOS.

Stage	5 Class Wake, N1, N2, N3, REM	4 Class Wake, Light, N3, REM (Light= N1+N2)	3 Class Wake, NREM, REM (NREM= N1+N2+N3)	2 Class Wake, Sleep (Sleep= N1+N2+ N3+REM)
Wake	0.84 (0.91) 0.41	0.88 (0.89)	0.88 0.89	0.89 (0.87)
N1	0.12 (0.63) 0.84	0.77 (0.85)		
N2	0.85 (0.76) 0.68		0.87 (0.91)	
N3	0.41 (0.41)	0.84 (0.74)		0.92 (0.93)
REM		0.71 (0.45)	0.68 (0.44)	
Accuracy	0.74	0.82	0.86	0.91
kappa	0.63	0.72	0.75	0.81

IV. CONCLUSIONS

We have examined the feasibility of automatically scoring sleep from the unobtrusive in-ear EEG from older adults. The analysis has shown a substantial agreement between the predicted stages and the gold standard manually scored stages (hypnogram). This makes the proposed method a viable alternative for sleep monitoring in older adults, paving the way for the use of Hearables in the community. A larger study involving a number of people living with dementia is underway.

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